Dependent Origination: A Mathematical Formalization of Interdependence in Quantum and Cosmological Systems

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We present a rigorous mathematical formalization of *dependent origination* (Pratītyasamuṭpāda), the principle taught by Buddha Shakyamuni, as a universal framework for causality in quantum and cosmological contexts. By excluding the alternatives of self-causation, external causation, dual causation, and causeless emergence, we show that only an interdependent-conditions model remains logically and empirically viable. We employ vacuum wave-functions, discrete-time Markov chains, and a fractal nesting of spacetime spheres (golden-ratio scaled) to capture probabilistic interdependence. Two TikZ diagrams—a directed acyclic graph and a fractal nesting—visualize the causal flow without disturbing text flow. Finally, we derive testable predictions for gravitational lensing, CMB anomalies, and LHC bounds.

Keywords: Dependent origination; Causality; Markov chains; Fractal geometry; Quantum cosmology

I. INTRODUCTION

Dependent origination posits that no phenomenon exists in isolation: each arises from mutually sustaining conditions. Rejecting self-causation, purely external causation, dual causation, and causeless emergence, we reinterpret this ancient principle in modern physics. Our framework integrates quantum vacuum fluctuations, Markov-chain transition dynamics, and golden-ratio fractal embeddings to yield a coherent, testable model.

II. MODELS OF CAUSATION

We consider four archetypal models:

1. *Self-Causation*: A phenomenon produces itself (infinite regress).
2. *External Causation*: A phenomenon is entirely sourced externally (violates relationality).
3. *Dual Causation*: Combines self and external causation (inconsistent).
4. *Non-Causation*: Spontaneous emergence (violates conservation).

Each fails logical or empirical consistency, leaving only an interdependent network.

III. FORMALISM

A. Vacuum Wave Function

Define

$$\Psi_{\text{void}} = \sum_i \Psi_i, \tag{1a}$$

$$\Psi_{\text{void}} = 0, \tag{1b}$$

where each Ψ_i is a vacuum fluctuation state (energy, field-mode, etc.). The sum-to-zero enforces overall conservation.

B. Markov-Chain Causality

Let $\{X_n\}$ be a discrete-time chain with

$$P(X_{n+1} = x \mid X_n = x_n, \ldots, X_0 = x_0) = P(X_{n+1} = x \mid X_n = x_n)\,. \tag{2}$$

The transition matrix is

$$P = \begin{pmatrix} P(F\to F) & P(F\to E) & 0 \\ 0 & P(E\to E) & P(E\to M) \\ 0 & 0 & P(M\to M) \end{pmatrix},$$

with states F=Fluctuation, E=Energy, M=Matter. It admits a unique steady state π satisfying $\pi P = \pi$.

C. Fractal Nesting & Golden Ratio

Embed spacetime as nested 3-spheres of radii:

$$R_{n+1} = R_n\,\phi, \tag{3a}$$

$$\phi = \frac{1+\sqrt{5}}{2}, \tag{3b}$$

with volumes $V_n \propto \phi^{-n}$ and energy densities

$$\rho_n = \frac{E_{\text{tot}} e^{-an}}{V_n}, \quad \sum_n E_n = E_{\text{tot}}.$$

Cosmic expansion is $R(t) = R_0 e^{Ht}$.

D. Energy–Time Uncertainty

Quantum fluctuations obey

$$\Delta E \, \Delta t \geq \hbar,$$

linking indeterminacy to temporal emergence.

IV. ILLUSTRATIONS

A. Directed Acyclic Graph

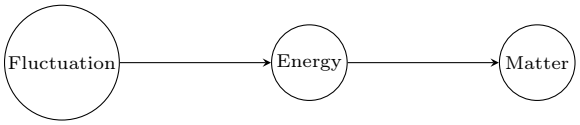


FIG. 1. Causal flow from vacuum fluctuations to energy to matter.

B. Fractal Nesting Diagram

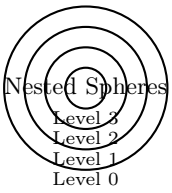


FIG. 2. Fractal nesting of spacetime spheres scaled by ϕ.

V. APPLICATIONS

A. Cosmology

Interdependent cycles predict enhanced cluster-scale lensing ($\approx 4\times$ standard) and CMB multipole anomalies for $\ell > 2000$.

B. Quantum Mechanics

Vacuum fluctuations $\rightarrow$ particle creation rates governed by $\Delta E \, \Delta t \geq \hbar$ under interdependence constraints.

VI. TESTABLE PREDICTIONS

- **Gravitational Lensing:** $\approx 4\times$ stronger effective gravity in clusters.
- **CMB Anomalies:** Excess power at $\ell > 2000$ beyond ΛCDM.
- **LHC Bounds:** Deviations in cross-sections from fractal-spacetime corrections.

VII. CONCLUSION

Dependent origination, formalized via vacuum wavefunctions, Markov chains, and golden-ratio fractals, yields a consistent, predictive framework unifying quantum and cosmological phenomena.

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